

LAKSHMI OMKARESWAR THUMMAGUNTA

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OBJECTIVE

Aspiring ECE student specializing in VLSI Design and Embedded Systems, passionate about RTL design, FPGA development, and real-world hardware innovation.

EDUCATION

B.E in Electronics and Communication Engineering, SCSVMV (Deemed to be University) 2024 – Present
Honours in VLSI Design Kanchipuram, TN
CGPA - 8.60/10.00

Diploma in Electronics and Communication Engineering, 2021 – 2024
Audisankara College of Engineering & Technology Nellore, AP

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Embedded C, HDL, VHDL, Verilog HDL, Digital Logic Design
- **Design:** FSM Design, RTL Design, Digital Electronics, FPGA Design, ASIC Design, UART/SPI/I2C
- **Embedded Systems:** ESP32, 8051 Microcontroller, Hardware Interfacing, Sensor Integration
- **Tools:** Vivado, LT-Spice, P-Spice
- **Domains:** VLSI Design, Embedded Systems, IoT Development, Wireless Communication

PROFESSIONAL EXPERIENCE

Internship Trainee, Andhra Pradesh Power Generation Corporation Limited 11/2023 – 05/2024
Nellore, Andhra Pradesh, India

- Gained hands-on experience in power plant operations, electrical systems, and thermal energy processes.
- Contributed to efficiency analysis, maintenance support, and data collection, developing a strong understanding of power generation systems.
- Collaborated with industry professionals to strengthen problem-solving abilities and gain practical exposure to power systems, energy management, and technical analysis.

PROJECTS

AI-Based Automatic Bike Cleaning System 11/2025 – 03/2026

- Designed FSM-based automated control system using sensor inputs and microcontroller.
- Implemented real-time hardware interfacing (motors, pumps, sensors) via GPIO.
- Optimized system efficiency (70% water saving, 2–3 min cycle).
- Finalist – **InnoWAH (PALS)**.

Mini RISC PROCESSOR 09/2025 – 10/2025

- Designed and simulated an 8-bit single-cycle RISC processor using Verilog HDL with ALU, register file, control unit, and memory interface in Vivado.

AI-Powered OLED Display Interface ESP32 04/2025 – 04/2025

- Created a cutting-edge embedded system that combines OpenAI API with a 128x64 OLED display, employing efficient I2C communication along with optimized memory and power management for enhanced edge-AI applications.

CERTIFICATIONS & RECOGNITIONS

VLSI Design – Internshala
Completed an 8-week VLSI Design training covering HDL fundamentals, combinational & sequential logic, FSMs, and FPGA based system design

Finalist – InnoWAH Challenge (PALS)
• **Selected as Finalist – InnoWAH (PALS)**, for designing a smart, sensor-driven automated system with real-world impact.

Special Prize for Innovative Engineering Solution - Think2Impact
• Successfully presented and validated the project in a **competitive, industry-oriented evaluation round**.